Router-on-a-Stick Inter-VLAN Routing Lab Guide

CCNA Practical Guide: Sub-Interfaces, 802.1Q Trunking, & Dynamic Refactoring

---

1. Theory & Technical Concepts

The VLAN Isolation Problem: By design, a Virtual Local Area Network (VLAN) creates an isolated broadcast domain at Layer 2. Devices inside one VLAN (e.g., VLAN 10) cannot communicate directly with devices in another VLAN (e.g., VLAN 20), even if connected to the same physical switch chassis. To pass traffic between distinct VLANs, packets must be routed at Layer 3 (Network Layer).

Router-on-a-Stick Overview: Router-on-a-Stick (RoAS) is a network configuration in which a single physical interface between a Layer 2 switch and a router is configured to carry traffic for multiple VLANs. The physical interface on the router is logically divided into multiple sub-interfaces, each assigned to a specific VLAN with 802.1Q encapsulation.

2. Practical Lab Tasks & Configuration

Task 1: VLAN Creation & Port Assignment

On the Layer 2 switch, create three VLANs named Insta (VLAN 10), Zomato (VLAN 20), and Flipkart (VLAN 30), and assign access ports to each connected host.

```
Switch# configure terminal

! Create VLANs
Switch(config)# vlan 10
Switch(config-vlan)# name Insta
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Zomato
Switch(config-vlan)# vlan 30
Switch(config-vlan)# name Flipkart
Switch(config-vlan)# exit

! Assign Access Ports
Switch(config)# interface FastEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit

Switch(config)# interface FastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

```
Switch(config)# interface FastEthernet0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
```

Task 2: Configure Router Sub-Interfaces

Bring up physical interface GigabitEthernet0/0 on the router and create logical sub-interfaces for each VLAN with corresponding 802.1Q encapsulation and default gateway IP addresses.

```
Router# configure terminal

! Bring up physical link
Router(config)# interface GigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)# exit

! Configure Sub-interface for VLAN 10 (Insta)
Router(config)# interface GigabitEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

! Configure Sub-interface for VLAN 20 (Zomato)
Router(config)# interface GigabitEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit

! Configure Sub-interface for VLAN 30 (Flipkart)
Router(config)# interface GigabitEthernet0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# exit
```

Task 3: Trunk Port Configuration & Verification

Configure the switch port connected to the router as an 802.1Q trunk port to multiplex traffic for all VLANs across the single physical link.

```
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit

! Verification Command
Switch# show interfaces trunk
```

Task 4: Connectivity Test & Ping Results

Assign host IPs: PC-Insta (192.168.10.10/24, GW: 192.168.10.1) and PC-Zomato (192.168.20.10/24, GW: 192.168.20.1). Execute ICMP ping test from PC-Insta to PC-Zomato.

```
C:\> ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:
Request timed out.                          <-- Initial ARP resolution at router
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
```

Technical Flow Analysis: 1. PC-Insta determines 192.168.20.10 is outside its subnet and sends the packet to gateway 192.168.10.1.
2. The switch tags the frame with VLAN 10 and sends it over the trunk link.
3. The router receives the frame on sub-interface Gi0/0.10, checks its routing table, and routes the packet out sub-interface Gi0/0.20 tagged as VLAN 20.
4. The switch receives the VLAN 20 frame and forwards it out access port Fa0/2 to PC-Zomato.

Task 5: Refactoring Configuration for 'Spotify' VLAN

Add a fourth VLAN called 'Spotify' (VLAN 40) non-disruptively to both the switch and router configuration without interrupting active communication on existing VLANs.

```
! 1. Switch Configuration
Switch# configure terminal
Switch(config)# vlan 40
Switch(config-vlan)# name Spotify
Switch(config-vlan)# exit
Switch(config)# interface FastEthernet0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# exit

! 2. Router Configuration
Router# configure terminal
Router(config)# interface GigabitEthernet0/0.40
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
Router(config-subif)# exit

! 3. Verification Commands
```

```
Router# show ip route connected
Switch# show vlan brief
```